Mathematics for Computing

Assignment 1

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Roadmap

1. Sets
2. Types of Sets
3. Set Builder Form
4. Inclusion–Exclusion
5. Partially Ordered Sets

What is a Set?

Definition

A set is a well-defined, unordered collection of distinct objects. The objects are called *elements* of the set.

Computing example

Let

$$D = \{\text{Python}, \text{Java}, \text{SQL}, \text{C}\}.$$

Here, each programming language occurs as one distinct element.

- Order does not change a set.
- Repeated elements are written only once.

Cardinality of a Set

The **cardinality** of a set is its number of distinct elements and is written as $|A|$.

Example

Consider the set of binary digits:

$$B = \{0, 1\}.$$

Therefore,

$$|B| = 2.$$

Important

If an element is repeated in a description of a set, it is still counted only once.

1. Finite Set

A finite set contains a limited number of distinct elements.

Data Science example

Suppose a classification model uses four labels:

$$C = \{\text{cat}, \text{dog}, \text{bird}, \text{fish}\}.$$

Then $|C| = 4$, so C is finite.

$$C = \{x \mid x \text{ is one of the four chosen class labels}\}.$$

2. Infinite Set

An infinite set has no finite number of elements.

Computing example

The set of possible real-valued weights of a machine-learning model can be represented by

$$W = \{x \mid x \in \mathbb{R}\}.$$

Since there are infinitely many real numbers, W is infinite.

3. Null (Empty) Set

An empty set contains no elements. It is denoted by $\varnothing$.

AI example

Suppose a search system is asked to return products satisfying an impossible combination of filters. If no product satisfies the query,

$$R = \varnothing.$$

Thus R is an empty set.

4. Countable Set

A set is countable when its elements can be listed one by one.

Computing example

The first six HTTP status codes in a selected list can be written as

$$S = \{200, 201, 204, 301, 400, 404\}.$$

They can be individually indexed, so S is countable.

$$S = \{x \mid x \text{ is one of the six selected status codes}\}.$$

5. Power Set

The power set $\mathcal{P}(A)$ is the set of all subsets of A.

Example

Let

$$A = \{\text{CPU}, \text{GPU}, \text{RAM}\}.$$

Since $|A| = 3$,

$$|\mathcal{P}(A)| = 2^3 = 8.$$

The power set contains $\varnothing$, the three one-element subsets, the three two-element subsets, and A itself.

6. Subset

If every element of A is also an element of B, then

$$A \subseteq B.$$

Data example

Let

$$A = \{\text{mean}, \text{median}\}, \qquad B = \{\text{mean}, \text{median}, \text{mode}\}.$$

Every element of A occurs in B, hence

$$A \subseteq B.$$

7. Equal Sets

Two sets are equal when they contain exactly the same elements.

Example

$$A = \{\text{red}, \text{green}, \text{blue}\}, \qquad B = \{\text{blue}, \text{red}, \text{green}\}.$$

The order is different, but the elements are identical. Therefore

$$A = B.$$

$$A = B \iff \forall x \, (x \in A \Leftrightarrow x \in B).$$

8. Equivalent Sets

Two sets are equivalent if they have the same cardinality.

Example

Let

$$A = \{10, 20, 30\}, \qquad B = \{\text{HTML}, \text{CSS}, \text{JavaScript}\}.$$

Both sets have three elements, so

$$|A| = |B| = 3.$$

Hence they are equivalent sets.

9. Superset

If $A \subseteq B$, then B is a superset of A, written as

$$B \supseteq A.$$

Example

Let

$$A = \{\text{train}, \text{test}\}, \qquad B = \{\text{train}, \text{validation}, \text{test}\}.$$

Then $A \subseteq B$, so

$$B \supseteq A.$$

10. Singleton Set

A singleton set contains exactly one element.

AI example

If a binary classifier produces one selected class for a particular record, the set of that record's predicted classes can be

$$P = \{\text{spam}\}.$$

Thus,

$$|P| = 1,$$

so P is a singleton set.

Set-Builder Form: Idea

Set-builder notation describes a set by stating a property that its elements must satisfy.

$$A = \{x \mid P(x)\}.$$

Example

The positive even integers below 12 are

$$A = \{2, 4, 6, 8, 10\}.$$

In set-builder form:

$$A = \{x \in \mathbb{N} \mid x \text{ is even and } x < 12\}.$$

Set-Builder Examples I

Q1. Multiples of 7 not exceeding 42

$$A = \{7, 14, 21, 28, 35, 42\}$$

$$A = \{x \mid x = 7n,\ 1 \leq n \leq 6,\ n \in \mathbb{N}\}.$$

Q2. Perfect squares from 1 through 81

$$B = \{1, 4, 9, 16, 25, 36, 49, 64, 81\}$$

$$B = \{x \mid x = n^2,\ 1 \leq n \leq 9,\ n \in \mathbb{N}\}.$$

Set-Builder Examples II

Q3. Integers between -5 and 8

$$C = \{x \mid x \in \mathbb{Z},\ -5 \le x \le 8\}.$$

Q4. Odd natural numbers below 20

$$D = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$$

$$D = \{x \mid x = 2n + 1,\ 0 \le n \le 9,\ n \in \mathbb{N}_0\}.$$

Set-Builder Examples III

Q5. Positive integers divisible by 4

$$E = \{4, 8, 12, 16, 20, \ldots\}$$

$$E = \{x \in \mathbb{N} \mid 4 \mid x\}.$$

Q6. Numbers leaving remainder 2 when divided by 5

$$F = \{2, 7, 12, 17, 22, \ldots\}$$

$$F = \{x \mid x = 5n + 2,\ n \in \mathbb{N}_0\}.$$

Set-Builder Examples IV

Q7. Factors of 36

$$G = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$$

$$G = \{x \in \mathbb{N} \mid x \mid 36\}.$$

Q8. Ordered pairs with sum 10

$$H = \{(1, 9), (2, 8), (3, 7), (4, 6), (5, 5), \ldots\}$$

$$H = \{(x, y) \in \mathbb{N}^2 \mid x + y = 10\}.$$

Set-Builder Examples V

Q9. Multiples of 9 below 60

$$I = \{9, 18, 27, 36, 45, 54\}$$

$$I = \{x \mid x = 9n,\ 1 \leq n \leq 6,\ n \in \mathbb{N}\}.$$

Q10. Integers whose absolute value is at most 3

$$J = \{-3, -2, -1, 0, 1, 2, 3\}$$

$$J = \{x \in \mathbb{Z} \mid |x| \leq 3\}.$$

Inclusion–Exclusion Principle

The Inclusion–Exclusion Principle counts distinct elements in a union while correcting for overlap.
For two sets:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

For three sets:

$$\begin{aligned} |A \cup B \cup C| = {} & |A| + |B| + |C| \\ & - |A \cap B| - |A \cap C| - |B \cap C| \\ & + |A \cap B \cap C|. \end{aligned}$$

Inclusion–Exclusion: Computing Example

In a class of 50 students:

- 28 know Python,
- 22 know Java,
- 12 know both.

The number who know at least one of the two languages is

$$|P \cup J| = 28 + 22 - 12 = 38.$$

So **38 students** know Python or Java (or both).

General Inclusion–Exclusion

For finite sets $A_1, \ldots, A_n$,

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i} |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots .$$

Pattern

Add individual sizes, subtract pairwise overlaps, add triple overlaps, and continue with alternating signs.

Partially Ordered Sets (Posets)

A partially ordered set is a set together with a relation satisfying:

1. **Reflexive:** $a \leq a$.
2. **Antisymmetric:** $a \leq b$ and $b \leq a$ imply $a = b$.
3. **Transitive:** $a \leq b$ and $b \leq c$ imply $a \leq c$.

A poset is commonly written as $(P, \leq)$.

Poset Example: Divisibility

Let

$$P = \{1, 2, 4, 8\}$$

and define

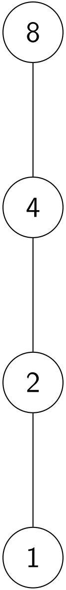
$$a \preceq b \iff a \mid b.$$

For example:

$$1 \mid 2, \qquad 2 \mid 4, \qquad 4 \mid 8, \qquad 1 \mid 8.$$

This relation is reflexive, antisymmetric, and transitive, so $(P, \preceq)$ is a poset.

Hasse Diagram: Divisibility Poset

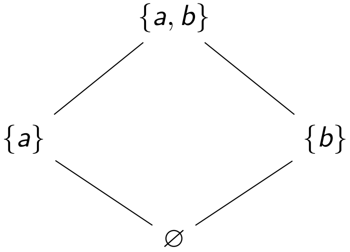


Hasse Diagram: Subset Poset

Consider

$$P = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}$$

ordered by $\subseteq$.



Summary

- Sets provide a basic way to represent collections of objects.
- Cardinality measures the number of distinct elements.
- Set-builder notation describes elements through properties.
- Inclusion–Exclusion corrects double counting in unions.
- Posets model relations that are reflexive, antisymmetric, and transitive.
- Hasse diagrams give a compact visual representation of finite posets.

Key idea

These concepts are useful for organizing data, describing relationships, and reasoning about discrete structures in computing.

Thank You

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